Transfer Entropy of a Round-Level Controlled q-Voter

Exact finite-population formulae and a closed mean-field approximation

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What this note contains. This report isolates the theoretical calculation only. We use the fixed classical rule agreed for the controlled imitation game: an ordinary focal agent follows a unanimity q-voter rule; during a controlled microscopic update, the controller occupies one of the same q social-input slots and advocates a fixed target opinion A. The controller senses and chooses its action once per population round. If it chooses advocacy, exactly b of the N microscopic update positions are selected uniformly in advance. From this specification we derive the microscopic kernels K_0, K_1, the whole-round kernels R_0, R_1, the exact transfer entropy, and a closed weak-control approximation. No microscopic reversibility assumption is used.

1 The model in one picture and all symbols

Plain-language picture. At the start of round k, the population contains some number n_k of agents supporting the target opinion A. The controller looks at only q_c randomly sampled agents, obtains a noisy measurement Y_k, and then chooses either NOOP or ADV. If it advocates, it preassigns exactly b of the N microscopic update opportunities to receive one controller message. The population then evolves for the whole round. We want to know how much information the controller action carries about the next population state once the current state is already known.

The round-level causal chain is

$$n_k \longrightarrow Y_k \longrightarrow U_k \longrightarrow \{N \text{ microscopic updates}\} \longrightarrow n_{k+1}. \tag{1}$$

We use the following notation throughout.

Symbol	Meaning
N	Population size. One round contains exactly N microscopic focal-update positions.
n_k	Number of agents supporting target opinion A at the beginning of round k; $n_k \in \{0, \ldots, N\}$.
$x_k = n_k/N$	Fraction of the population supporting A.
q	Number of social-input slots seen by one focal agent.
q_c	Number of agents sampled by the controller once at the beginning of the round.
$Y_k = y$	Number of sampled agents supporting A, so $y \in \{0, \ldots, q_c\}$.
U_k	Controller action, $U_k \in \{\text{NOOP}, \text{ADV}\}$.
θ	Soft-controller threshold in target fraction.
β	Soft-controller slope: larger β makes the action closer to a hard threshold.
b	Exact number of controlled microscopic positions in an advocacy round.
$c = b/N$	Fraction of microscopic update positions controlled in an advocacy round.
K_0	One-microscopic-update transition kernel under an ordinary social update.
K_1	One-microscopic-update transition kernel when one of the q inputs is controller advocacy for A.
R_0	Whole-round transition kernel under NOOP.
R_1	Whole-round transition kernel under ADV with exactly b controlled positions.
a_n	Marginal advocacy probability $\mathbb{P}(U_k = \text{ADV} \mid n_k = n)$ after averaging over sensor noise.
$T(n)$	State-local transfer entropy at current state n.
$\mathcal{T}_k$	Transfer entropy averaged over the distribution of states visited at round k.

The classical motivation is the nonlinear q-voter herding mechanism: an agent changes opinion when confronted with q agents holding the opposite opinion. Irisarri et al. embed that mechanism in a reversible herding/anticonformity model; here we retain the herding-style unanimity rule but deliberately do not impose their reversibility construction [1].

2 Step 1: eliminate the noisy sensor and obtain the action probability

What we calculate. The controller does not know n exactly. It samples q_c agents and applies the soft policy to the observed target fraction y/q_c. The first useful simplification is to average over every possible sensor result. This gives a direct formula for the probability of advocacy conditioned only on the true population state n.

Sampling is without replacement, so the measurement is hypergeometric:

$$\boxed{S(y \mid n) = \mathbb{P}(Y_k = y \mid n_k = n) = \frac{\binom{n}{y}\binom{N-n}{q_c-y}}{\binom{N}{q_c}}.} \tag{2}$$

The soft controller is

$$\boxed{\pi_1(y) = \mathbb{P}(U_k = \mathrm{ADV} \mid Y_k = y) = \sigma\left[\beta\left(\theta - \frac{y}{q_c}\right)\right], \qquad \sigma(z) = \frac{1}{1+e^{-z}}.} \tag{3}$$

Averaging Eq. (3) over Eq. (2) yields

$$\boxed{a_n = \mathbb{P}(U_k = \mathrm{ADV} \mid n_k = n) = \sum_{y=0}^{q_c} \frac{\binom{n}{y}\binom{N-n}{q_c-y}}{\binom{N}{q_c}} \sigma\left[\beta\left(\theta - \frac{y}{q_c}\right)\right].} \tag{4}$$

This is already a closed finite expression: there is no unspecified controller probability left.

A particularly useful exact simplification: $q_c = 2$, $\theta = 1/2$

Let

$$s = \sigma(\beta/2).$$

For $y = 0, 1, 2$, the advocacy probabilities are $s, 1/2, 1-s$, which are exactly linear in y. Since $\mathbb{E}[Y \mid n] = 2n/N$, Eq. (4) collapses to

$$\boxed{a_n = s + (1-2s)\frac{n}{N}.} \tag{5}$$

Writing $x = n/N$ and using $2\sigma(z) - 1 = \tanh(z/2)$,

$$\boxed{a(x) = \frac{1}{2} + \frac{1-2x}{2}\tanh\left(\frac{\beta}{4}\right).} \tag{6}$$

Thus for the common midpoint threshold and $q_c = 2$, the noisy sensor plus logistic controller becomes an *exact affine function of the true target fraction.*

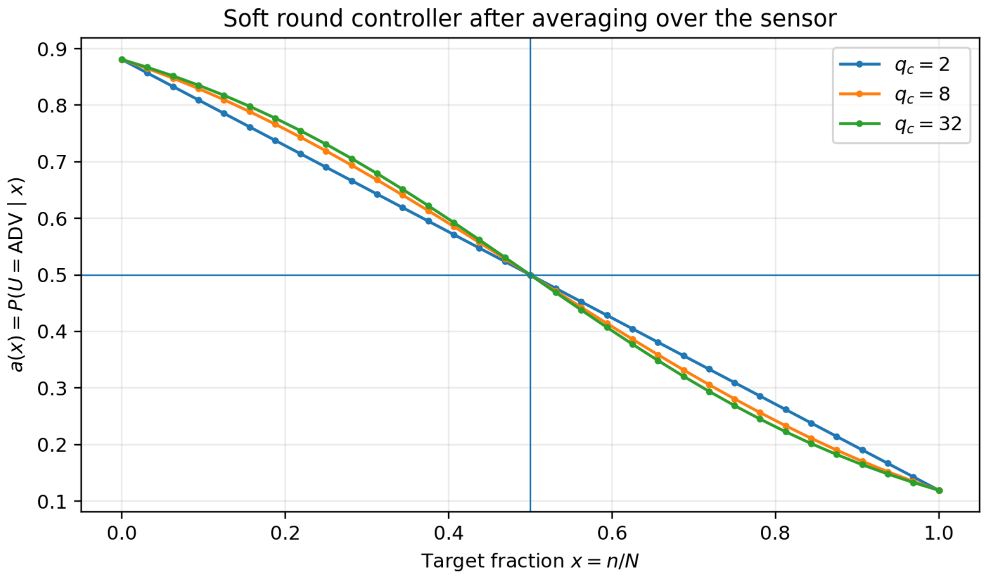

Figure 1: The exact advocacy probability after integrating out sensor noise for $N = 32$, $\theta = 1/2$, and an illustrative $\beta = 4$. The formulas are general; $\beta = 4$ is used only to make the theoretical curves concrete. Larger q_c sharpens the relation between the true state and the sampled state, while all curves pass through $a(1/2) = 1/2$.

3 Step 2: derive the microscopic transition kernels K_0 and K_1

What we calculate. During one microscopic update, only one focal agent can change opinion. Therefore the target headcount n can only move to $n + 1$, remain at n, or move to $n - 1$. We calculate these probabilities exactly for an ordinary update and for a controlled update.

Ordinary update K_0

A $B \to A$ switch requires choosing a B focal and then drawing q peers that are all A. Therefore

$$\boxed{K_0(n+1 \mid n) = \frac{N-n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}} \equiv p_0^+(n).} \tag{7}$$

Similarly, an $A \to B$ switch requires an A focal followed by q unanimously B peers:

$$\boxed{K_0(n-1 \mid n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}} \equiv p_0^-(n).} \tag{8}$$

The remaining probability is

$$\boxed{K_0(n \mid n) = 1 - p_0^+(n) - p_0^-(n).} \tag{9}$$

We use the convention $\binom{r}{q} = 0$ when $r < q$.

Controlled update K_1

At a controlled position, one input slot is already occupied by the controller saying A. A B focal now needs only the remaining $q - 1$ ordinary peers to support A:

$$\boxed{K_1(n+1 \mid n) = \frac{N-n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}} \equiv p_1^+(n).} \tag{10}$$

An A focal can never see q unanimous B inputs because the controller contributes one A input, so

$$\boxed{K_1(n-1 \mid n) = 0, \qquad K_1(n \mid n) = 1 - p_1^+(n).} \tag{11}$$

This asymmetry is intentional: the controller advocates only A. Transfer entropy requires a stochastic transition law, not microscopic reversibility.

4 Step 3: construct the whole-round kernels R_0 and R_1

What we calculate. Transfer entropy is evaluated on the round clock, but K_0 and K_1 describe only one focal update. We therefore combine N microscopic updates into one round-level transition law. Under NOOP every position is ordinary. Under ADV exactly b positions are controlled, chosen uniformly before the round begins.

If the controller does nothing, the round kernel is simply

$$\boxed{R_0 = K_0^N.} \tag{12}$$

For advocacy, the order of ordinary and controlled updates matters because the population state changes during the round. Let $F_{r,j}$ be the transition matrix after r positions when exactly j controlled positions have already occurred, averaged over all compatible preallocated schedules. Initialize

$$F_{0,0} = I.$$

At position $r+1$, the probability that the next preallocated slot is controlled is $(b-j)/(N-r)$; the probability it is ordinary is $[N-b-r+j]/(N-r)$. Hence

$$F_{r+1,j+1} \mathrel{+}= \frac{b-j}{N-r}\, F_{r,j}K_1, \tag{13}$$

$$F_{r+1,j} \mathrel{+}= \frac{N-b-r+j}{N-r}\, F_{r,j}K_0. \tag{14}$$

After all N positions,

$$\boxed{R_1 = F_{N,b}.} \tag{15}$$

This is exact and averages over all $\binom{N}{b}$ possible schedules without enumerating them.

For the special case $b = 1$, the exact result is especially transparent:

$$\boxed{R_1^{(b=1)} = \frac{1}{N}\sum_{r=0}^{N-1} K_0^r K_1 K_0^{N-1-r}.} \tag{16}$$

It is simply the average over the possible locations of the single controlled microscopic update.

5 Step 4: exact finite-N transfer entropy

What we calculate. At a fixed current state n, the only uncertainty about which round dynamics is used is the controller action. If NOOP and ADV produce nearly the same distribution of n_{k+1}, the action tells us almost nothing about the next state. If the two round kernels are very different, knowing the action is informative. This is exactly what transfer entropy measures after conditioning on the current state.

Define the action-unobserved next-state distribution at current state n:

$$\boxed{M_n(m) = [1-a_n]R_0(m \mid n) + a_n R_1(m \mid n).} \tag{17}$$

The exact state-local transfer entropy is

$$\boxed{\begin{aligned} T(n) = [1-a_n] \sum_m R_0(m \mid n) \log_2 \frac{R_0(m \mid n)}{M_n(m)} \\ + a_n \sum_m R_1(m \mid n) \log_2 \frac{R_1(m \mid n)}{M_n(m)}. \end{aligned}} \tag{18}$$

Equivalently,

$$\boxed{T(n) = \mathrm{JS}_{a_n}\left(R_0(\cdot \mid n), R_1(\cdot \mid n)\right),} \tag{19}$$

namely a weighted Jensen–Shannon divergence between the two possible round transition laws.

This is the exact finite-population answer once $N, q, q_c, b, \beta, \theta$ are specified. It contains no empirical MI estimator and requires no Monte Carlo simulation.

The binary action immediately gives the information bound

$$\boxed{0 \le T(n) \le h_2(a_n),} \qquad h_2(a) = -a\log_2 a - (1-a)\log_2(1-a). \tag{20}$$

To obtain the transfer entropy for an actual ensemble of rounds, let $P_k(n) = \mathbb{P}(n_k = n)$. The closed-loop round kernel is

$$R(m \mid n) = [1 - a_n]R_0(m \mid n) + a_n R_1(m \mid n), \tag{21}$$

so

$$P_{k+1}(m) = \sum_n P_k(n)R(m \mid n), \tag{22}$$

and

$$\boxed{\mathcal{T}_k = I(U_k; n_{k+1} \mid n_k) = \sum_n P_k(n)T(n).} \tag{23}$$

Why there is not one universal scalar TE without one more input. Equation (18) is closed at every state n. A single round-averaged number $\mathcal{T}_k$ additionally depends on how often the process visits each state, i.e. on $P_k(n)$ or an initial distribution P_0. In the pure unanimity q-voter, consensus states are absorbing; once the system is frozen, $R_0 = R_1$ at consensus and the long-time TE is zero. Therefore the state-local $T(n)$ and finite-horizon $\mathcal{T}_k$ are the natural theoretical quantities.

Horowitz and Sandberg emphasize transfer entropy as a directional measure of how transition probabilities are influenced by another process; in their linear Gaussian feedback model they can further reduce the general information expression to an explicit parameter formula [2]. The calculation below plays the analogous role for this discrete controlled imitation model.

6 Step 5: a true second-order closed approximation

What we calculate. The exact formula still requires the finite matrices R_0 and R_1. When advocacy changes the round transition law only moderately, we can expand the Jensen–Shannon divergence. This yields a compact approximation and shows explicitly how TE scales with control strength.

Write

$$R_1(m \mid n) = R_0(m \mid n) + \delta_n(m), \qquad \sum_m \delta_n(m) = 0.$$

Expanding Eq. (18) to second order in δ_n gives

$$\boxed{T(n) \simeq \frac{a_n(1-a_n)}{2\ln 2} \sum_{m: R_0(m|n)>0} \frac{\delta_n(m)^2}{R_0(m \mid n)}.} \tag{24}$$

The approximation has three immediately interpretable factors:

1. $a_n(1-a_n)$: the controller must have both actions available at the same state; it is maximal at $a_n = 1/2$.
2. δ_n^2: TE is second order in the separation between controlled and uncontrolled dynamics.
3. $1/R_0$: changes are especially informative where they move probability into outcomes that were unlikely under no control.

Equation (24) also explains why the approximation fails when control opens support that was impossible under NOOP. That occurs, for example, for $q = 1$ at $n = 0$: no-control dynamics stay at zero, while controlled updates can create the first A supporter. There the exact TE can be large and the small-perturbation expansion is not appropriate.

7 Step 6: closed mean-field TE as a function of $N, q, b, q_c, \beta, \theta, x$

What we calculate. We now remove the round matrices as well. For large N we approximate the population by the continuous fraction $x = n/N$. We calculate the mean drift and the intrinsic finite-population noise of an ordinary update, calculate how control changes the drift, and insert the signal-to-noise separation into the second-order information formula.

For large N and fixed q, the ordinary microscopic transition probabilities become

$$p_0^+(x) \simeq (1-x)x^q, \qquad p_0^-(x) \simeq x(1-x)^q. \tag{25}$$

The ordinary drift is

$$\boxed{f_0(x) = (1-x)x^q - x(1-x)^q.} \tag{26}$$

At a controlled microscopic position,

$$\boxed{f_1(x) = (1-x)x^{q-1},} \tag{27}$$

because $A \to B$ is blocked and $B \to A$ needs only $q-1$ ordinary A peers.

The actuation advantage of one controlled slot is therefore

$$\boxed{\Delta f_q(x) = f_1(x) - f_0(x) = x^{q-1}(1-x)^2 + x(1-x)^q.} \tag{28}$$

If a fraction $c = b/N$ of the round is controlled, the leading separation of the two round means is

$$\boxed{\Delta\mu(x) \simeq c\,\Delta f_q(x).} \tag{29}$$

The microscopic ordinary-update variance in the headcount increment is

$$\boxed{\nu_0(x) = p_0^+(x) + p_0^-(x) - f_0(x)^2.} \tag{30}$$

Freezing x across one sweep gives a round variance of the fraction approximately $\nu_0(x)/N$. Combining this with the second-order Jensen–Shannon expansion yields

$$\boxed{T_{\mathrm{MF}}(x) \simeq \frac{a(x)[1-a(x)]}{2\ln 2}\,\frac{Nc^2[\Delta f_q(x)]^2}{\nu_0(x)}.} \tag{31}$$

All controller parameters enter through $a(x)$ from Eq. (4); for $q_c = 2, \theta = 1/2$, substitute the explicit Eq. (6).

Closed scaling result. Because $c = b/N$,

$$Nc^2 = \frac{b^2}{N}.$$

Thus, in the weak-separation regime,

$$\boxed{T_{\mathrm{MF}}(x) \propto \frac{b^2}{N}.}$$

A fixed absolute actuation budget b becomes information-theoretically weaker as the population grows. To maintain an $O(1)$ local TE before saturation, the approximation suggests the crossover scaling $b \sim \sqrt{N}$. If instead a fixed fraction c is controlled, the action-conditioned distributions separate increasingly strongly with N until the exact TE approaches its binary-action ceiling $h_2(a(x))$; Eq. (31) should not be extrapolated beyond that overlap regime.

The same drift calculation gives a closed local control threshold. For $q \geq 2$ and $0 < x < 1/2$, an advocacy round reverses the negative ordinary drift when

$$\boxed{c > c_\star(x;q) = \frac{(1-x)^{q-1} - x^{q-1}}{(1-x)\left[x^{q-2} + (1-x)^{q-2}\right]}.} \tag{32}$$

This is a control threshold for mean drift, not a transfer-entropy threshold; the two answer different questions.

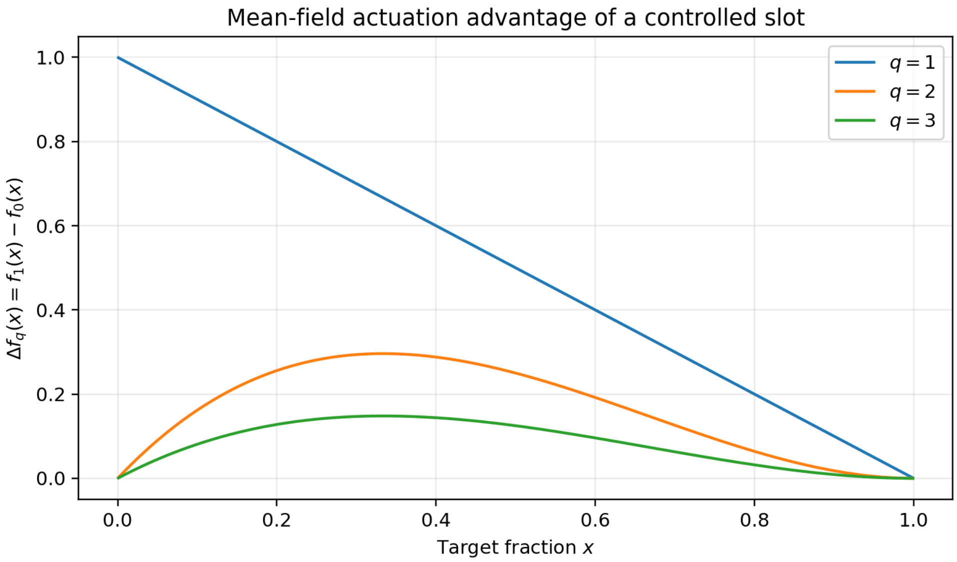

Figure 2: The mean-field actuation advantage $\Delta f_q(x)$ from Eq. (28). For larger q, one-slot advocacy becomes ineffective when the target is very rare because the remaining $q-1$ ordinary inputs must already contain target supporters.

8 Step 7: the special case $q = 1$ is analytically stronger

What is special about $q = 1$. An ordinary one-peer voter has zero mean drift: copying a random A is exactly balanced by copying a random B. A controlled position, however, gives the focal agent the controller's A input directly. This lets us calculate the expected effect of an entire advocacy round exactly, without a mean-field approximation.

For $q = 1$,

$$f_0(x) = 0, \qquad f_1(x) = 1 - x.$$

An ordinary microscopic update preserves $\mathbb{E}[x]$. A controlled microscopic update maps the current expected fraction according to

$$\mathbb{E}[x] \longmapsto \mathbb{E}[x] + \frac{1 - \mathbb{E}[x]}{N}.$$

Applying this exactly b times, regardless of where those b positions appear in the round, gives

$$\boxed{\mathbb{E}[x_{k+1} \mid x_k = x, \mathrm{ADV}] = 1 - (1-x)\left(1 - \frac{1}{N}\right)^b.} \tag{33}$$

Under NOOP,

$$\boxed{\mathbb{E}[x_{k+1} \mid x_k = x, \mathrm{NOOP}] = x.} \tag{34}$$

Therefore the exact mean separation is

$$\boxed{\Delta\mu_{q=1}(x) = (1-x)\left[1 - \left(1 - \frac{1}{N}\right)^b\right].} \tag{35}$$

If $b = cN$ and $N \to \infty$,

$$\boxed{\Delta\mu_{q=1}(x) \longrightarrow (1-x)(1-e^{-c}).} \tag{36}$$

This produces a sharp qualitative distinction:

$$\boxed{q = 1: \; T(0) > 0 \text{ for } b > 0, \qquad q \geq 2: \; T(0) = 0.} \tag{37}$$

For $q = 1$, advocacy can create the first target supporter from an all-B population. For $q \geq 2$, one controller input is insufficient: at least $q - 1$ ordinary target supporters are still required, so both R_0 and R_1 leave $n = 0$ unchanged.

9 Exact theoretical curves for $N = 32$

How to read the following plots. These are not simulation results. They are obtained by constructing K_0, K_1, computing the exact round kernels R_0, R_1 through the finite-b recursion, and evaluating Eq. (18). To make the curves concrete we set $N = 32$, $q_c = 2$, $\theta = 1/2$, and use illustrative softness $\beta = 4$. The round budget b and social order q are varied explicitly.

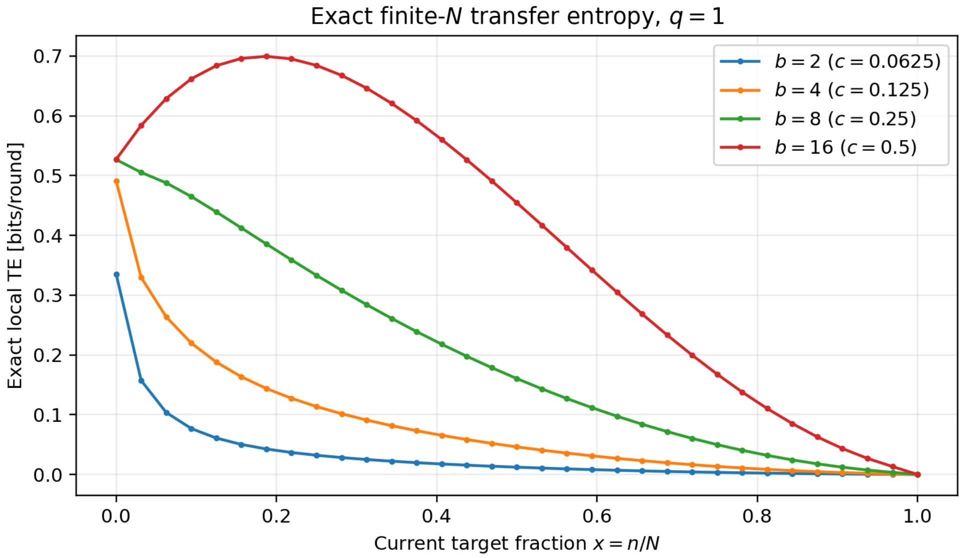

Figure 3: Exact state-local transfer entropy for $q = 1$. The large value near $x = 0$ is the seeding effect of Eq. (37): NOOP cannot leave the all-B state, while advocacy can. As b grows, the two round kernels become more distinguishable and TE approaches the action-entropy ceiling in parts of the state space.

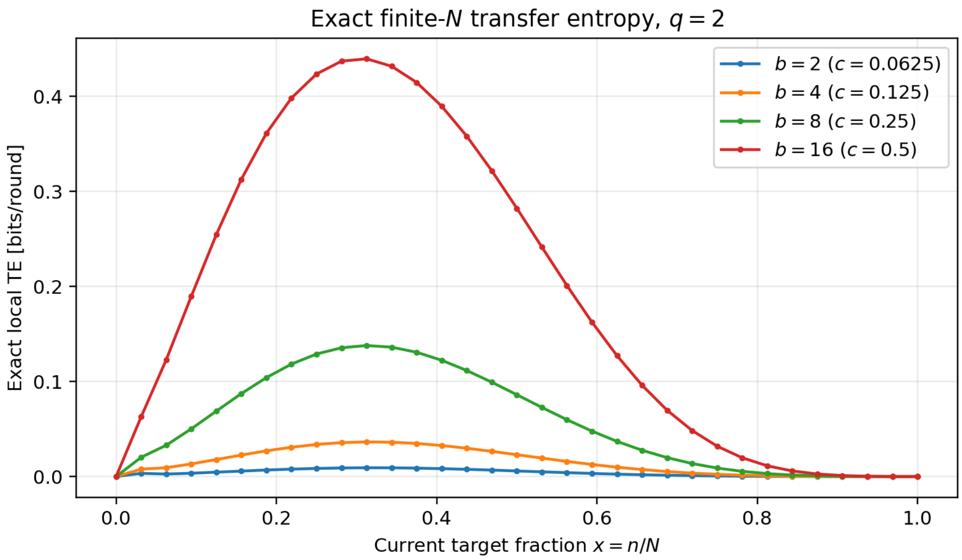

Figure 4: Exact state-local transfer entropy for $q = 2$. In contrast to $q = 1$, TE is exactly zero at $x = 0$ because the controller cannot create the first target supporter by itself. The information channel peaks at an interior target fraction, where advocacy can change the round dynamics but the population is not yet close to absorption.

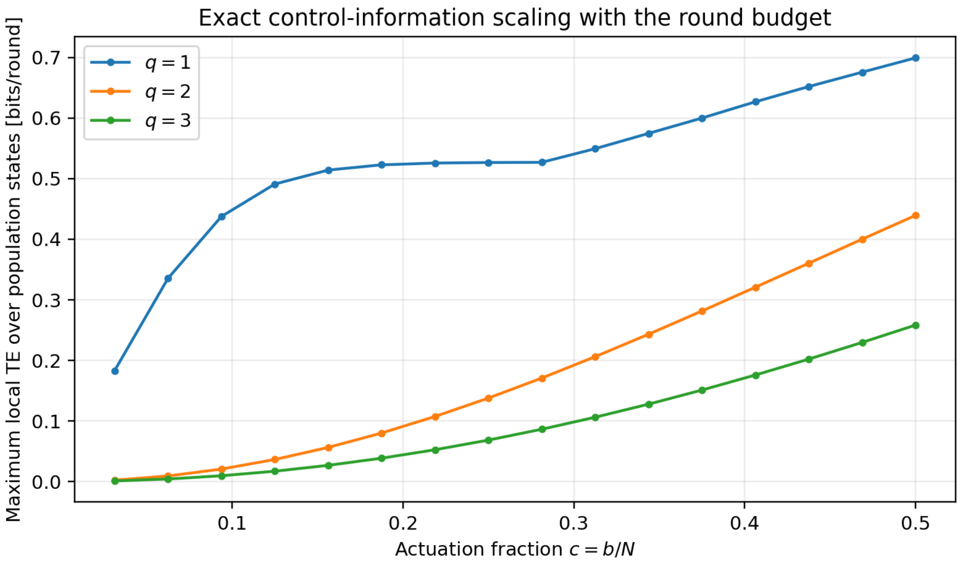

Figure 5: Maximum exact local TE over all population states as a function of $c = b/N$. For $q = 2$ and $q = 3$, the small-budget regime is approximately quadratic in c, as predicted by Eq. (31). The $q = 1$ curve departs from this simple regime early because control opens otherwise inaccessible transitions near the boundary.

For example, at $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$, and $b = 8$ ($c = 1/4$), the exact maximum state-local TE is approximately

$$0.527 \text{ bits/round for } q = 1, \qquad 0.138 \text{ bits/round for } q = 2, \qquad 0.068 \text{ bits/round for } q = 3.$$

The maxima for $q = 2$ and $q = 3$ occur at interior target fractions near $x \simeq 0.31$ and $x \simeq 0.34$, respectively.

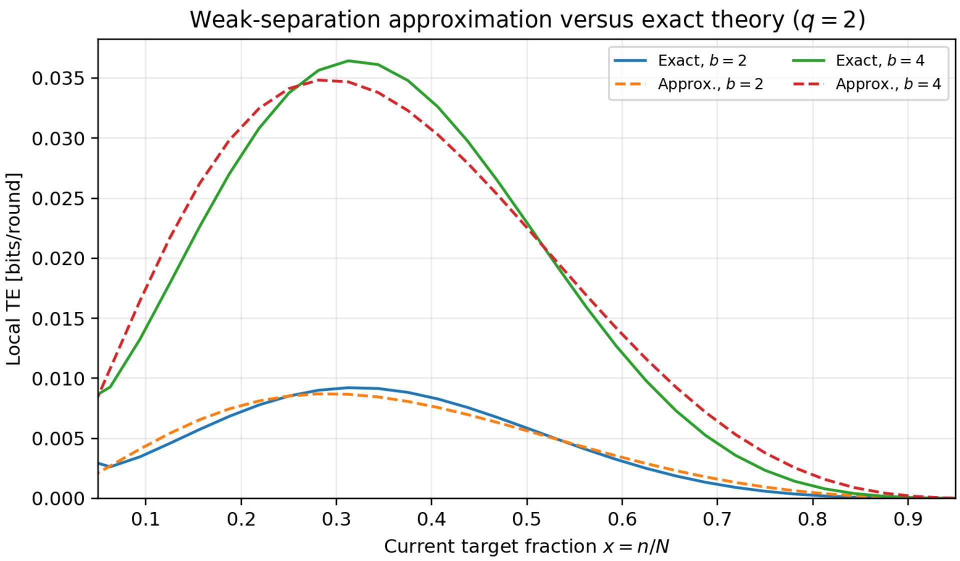

Figure 6: Exact $q = 2$ transfer entropy versus the closed weak-separation approximation in Eq. (31). For the small budgets $b = 2$ and $b = 4$, the approximation reproduces the shape and magnitude well across the interior state space. Over $x \in [0.0625, 0.9375]$, the RMS errors are about 4.5×10^{-4} and 1.8×10^{-3} bits/round, respectively.

10 The theoretical results, distilled

1. **The exact finite-N TE is available.** For every current headcount n, it is the weighted Jensen–Shannon divergence in Eq. (19) between the exact NOOP and ADV round kernels. The only averaging needed for a global $\mathcal{T}_k$ is over the state occupancy $P_k(n)$.
2. **The soft controller can be eliminated analytically.** Its exact state-conditioned advocacy probability is the hypergeometric sum in Eq. (4); for $q_c = 2, \theta = 1/2$ it becomes the simple affine function in Eq. (6).
3. **Weak-control TE has a closed parameter formula.** In the large-N, overlapping-kernel regime,

$$T_{\mathrm{MF}}(x) \simeq \frac{a(x)[1-a(x)]}{2\ln 2} \frac{N(b/N)^2[\Delta f_q(x)]^2}{\nu_0(x)}.$$

   This separates action entropy, actuation strength, and population noise.
4. **The budget scaling is nontrivial.** At fixed absolute b, the weak-control formula scales as b^2/N; an $O(1)$ information channel therefore requires roughly $b \sim \sqrt{N}$ before saturation effects become important. At fixed controlled fraction c, finite-population noise shrinks with N and TE moves toward its binary-action ceiling.
5. $q = 1$ **and** $q \geq 2$ **are qualitatively different at the boundary.** A one-slot controller can seed the target from $x = 0$ when $q = 1$, but cannot do so for $q \geq 2$. Consequently $T(0) > 0$ for $q = 1$ and $T(0) = 0$ for $q \geq 2$.
6. **Absorbing consensus explains vanishing asymptotic TE.** In the pure unanimity model, once consensus is reached the two actions no longer generate distinguishable next-state distributions. Therefore the meaningful theoretical quantities are state-local and finite-horizon transfer entropies rather than a nonzero stationary rate.

References

[1] L. Irisarri, L. Trigal, R. Toral, and G. Manzano, *Stochastic Thermodynamics of Social Imitation beyond Energetics*, arXiv:2511.14006v2, 2026.

[2] J. M. Horowitz and H. Sandberg, *Second-law-like inequalities with information and their interpretations*, arXiv:1409.5351, 2014.